
AWS
re:Invent
DECEMBER 1 – 5, 2025 | LAS VEGAS, NV

CON354

Amazon EKS Auto Mode: Evolving Kubernetes ops to enable innovation

Sai Vennam

he/him

Principal Containers Specialist

Amazon Web Services

Alex Kestner

he/him

Principal Product Manager – Technical

Amazon Web Services

Daniel Levine

he/him

Senior Lead Software Engineer

Capital One

From zero to hero in 5 minutes with EKS Auto Mode

Kubernetes and Amazon EKS

Kubernetes – the production platform



93%

Using in production or actively evaluating

© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.

CNCF Annual Survey 2024, cncf.io/reports/cncf-annual-survey-2024/

Why Kubernetes?



Simplicity

195 CNCF projects



Consistency

100s of compatible tools



Extensibility

Unlimited customization

aws

© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.



Amazon EKS

- The most trusted and secure way to run Kubernetes
- Enables you to build reliable, stable, and secure applications in any environment
- Fully upstream and certified conformant Kubernetes

aws

© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.

8 years of managed Kubernetes on AWS

2018

- EKS Generally Available
- Managed Cluster Version Updates
- GPU Support
- HIPAA eligible

2019

- ISO, PCI, and SOC Compliance
- Expansion to 15 AWS regions
- CSI drivers for EBS, EFS, FSx for Lustre
- Pod security policies
- Managed Node Groups
- EKS Fargate
- EKS on AWS Outposts

2020

- Price reduced to $0.10 per hour
- Secrets Encryption
- SLA raised to 99.95%
- EKS CIS Benchmark
- ACK Project
- Load balancer controller
- EKS on AWS Local zones
- K8s Resources in EKS console
- EKS Add-ons

2021

- OIDC access authentication
- Karpenter project
- Cluster creation reduced by 40%
- Control plane scaling
- EKS Anywhere GA
- EKS Connector
- FedRamp High Compliance
- EFA, P4d instance support

2022

- IPv6 clusters available
- PrivateLink support
- Local clusters on Outposts
- Trainium instances
- Add-ons from AWS marketplace
- Nitro enclaves
- Launch time reductions

2023

- Expansion to 32 AWS regions
- Kubernetes Network Policy enforcement
- Upgrade Insights
- Extended version support
- Version launch acceleration
- Pod Identity
- S3 Mountpoints CSI Driver

2024

- Amazon Linux 2023
- Automatic Version Upgrades
- Zonal Shift
- Full IPv6 support
- Multi-cluster dashboard
- Auto Mode
- Hybrid Nodes
- Node health & auto-repair

2025

- 100,000 node clusters
- Provisioned Control plane
- MCP Servers
- Managed Capabilities
- EKS Dashboard
- Default K8s API encryption
- Auto Mode Gov Cloud
- EKS Community Addons

aws

© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.

Launch Counter // 0



AMAZON EKS

Runs tens of millions

of clusters every year

© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.



AMAZON EKS

Auto Mode

Automate your entire Kubernetes cluster infrastructure

aws

© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.



WHAT IS AMAZON EKS

Auto Mode

- Application-ready clusters, with essential Kubernetes capabilities
- Dynamic compute scaling, AWS-managed instances, secure and healthy by default
- Automatic compute optimization, simplified cluster upgrades

aws

© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.



WHY AMAZON EKS

Auto Mode

- Accelerate innovation by offloading cluster operations to AWS
- Improve performance, availability, and security of your applications with AWS operational excellence
- Optimize compute costs with automatic capacity planning, maintenance and dynamic scaling

aws

© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.

Deploying and operating Kubernetes applications



Provision control plane



Install plug-ins



Select and provision compute



Deploy application



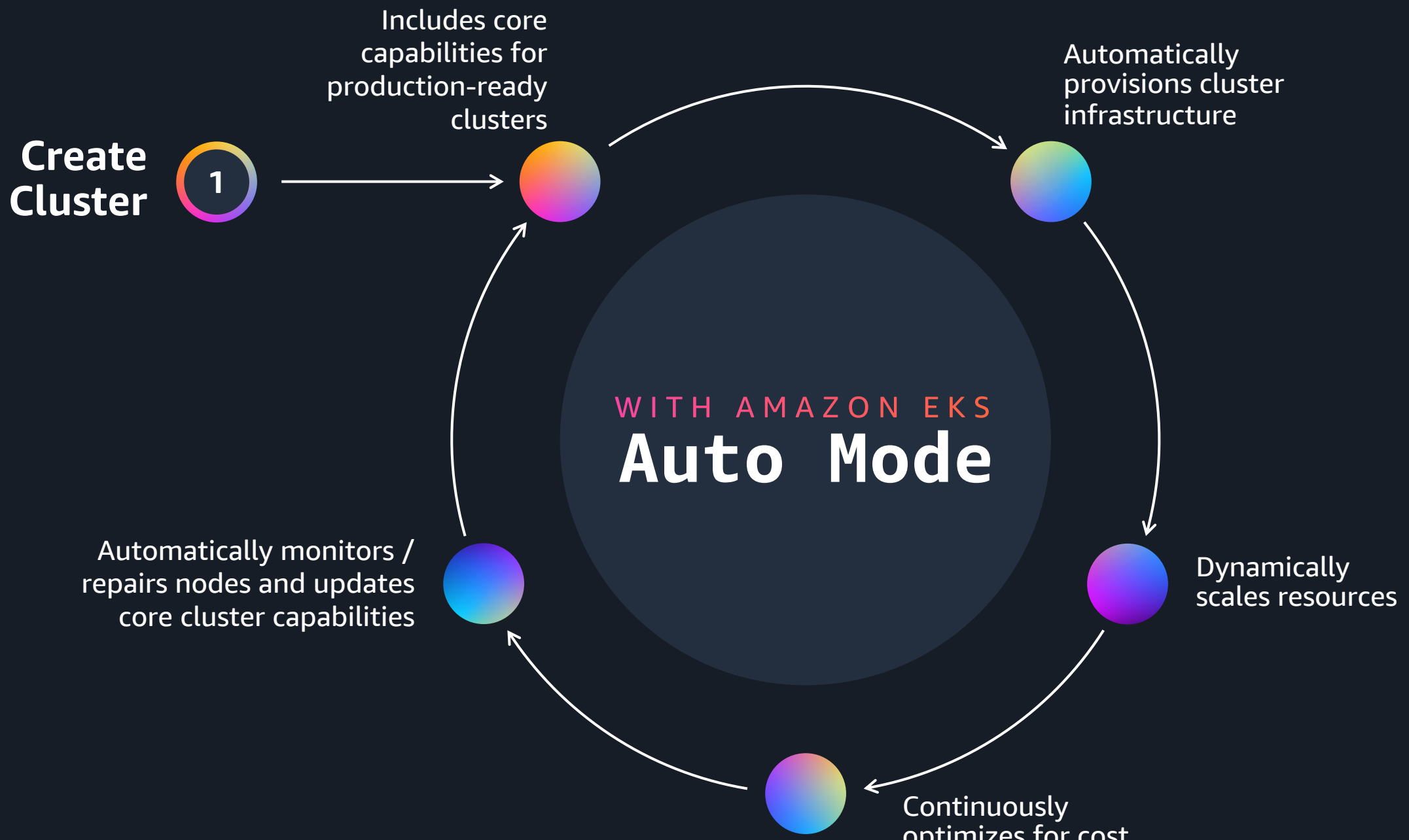
Monitor, repair, and upgrade



Analyze and optimize

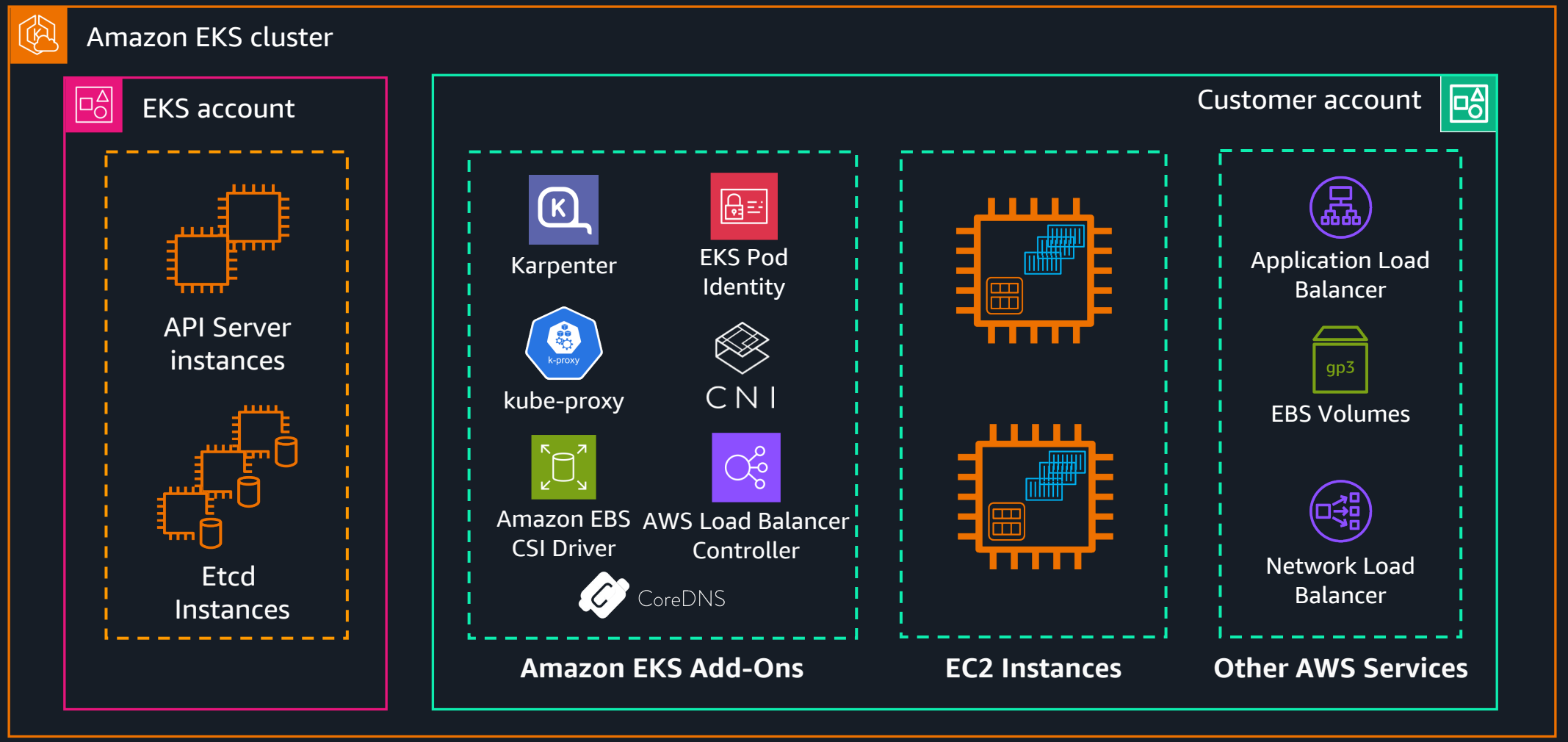
aws

© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.



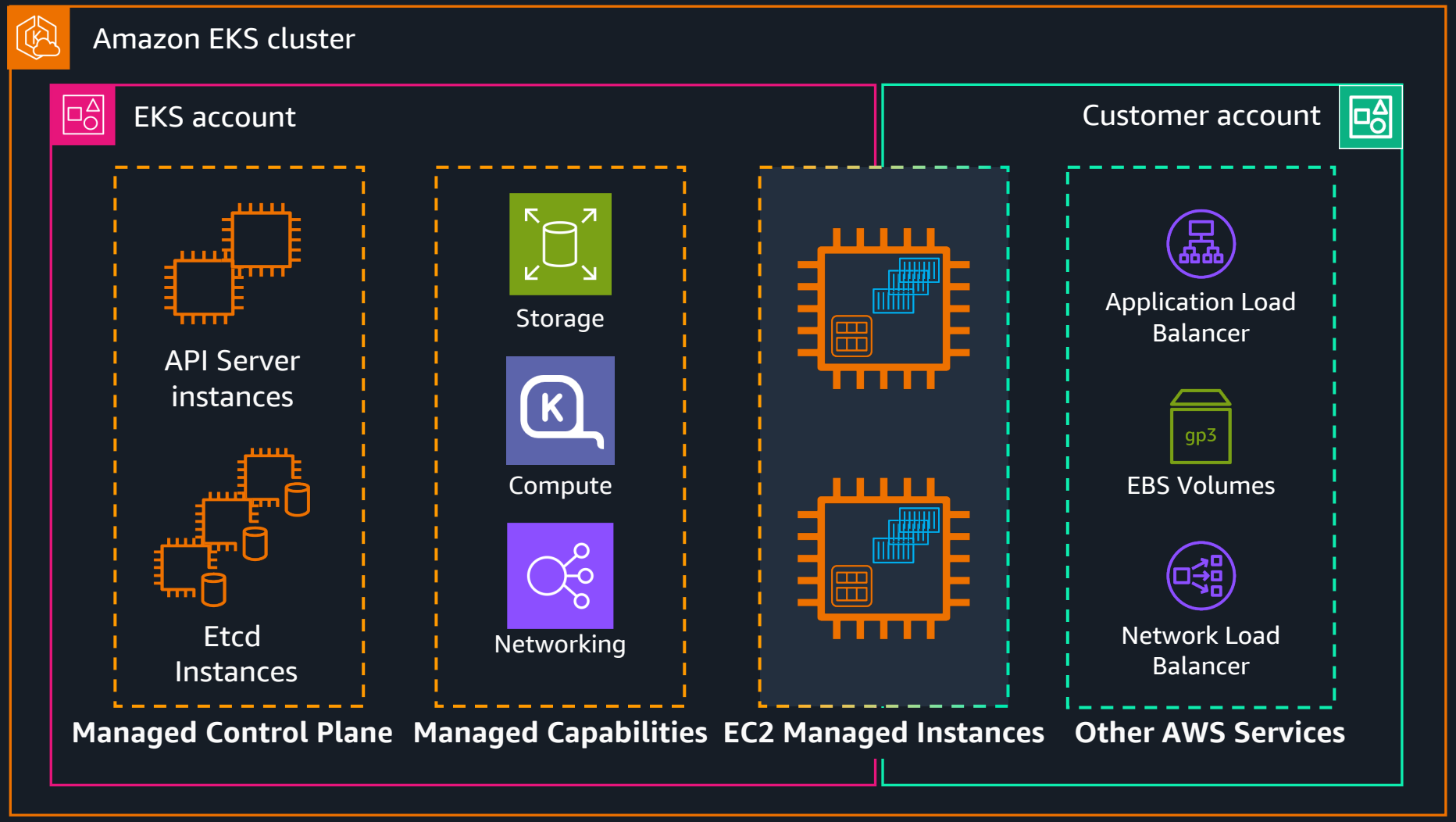
© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.

Standard Amazon EKS cluster architecture



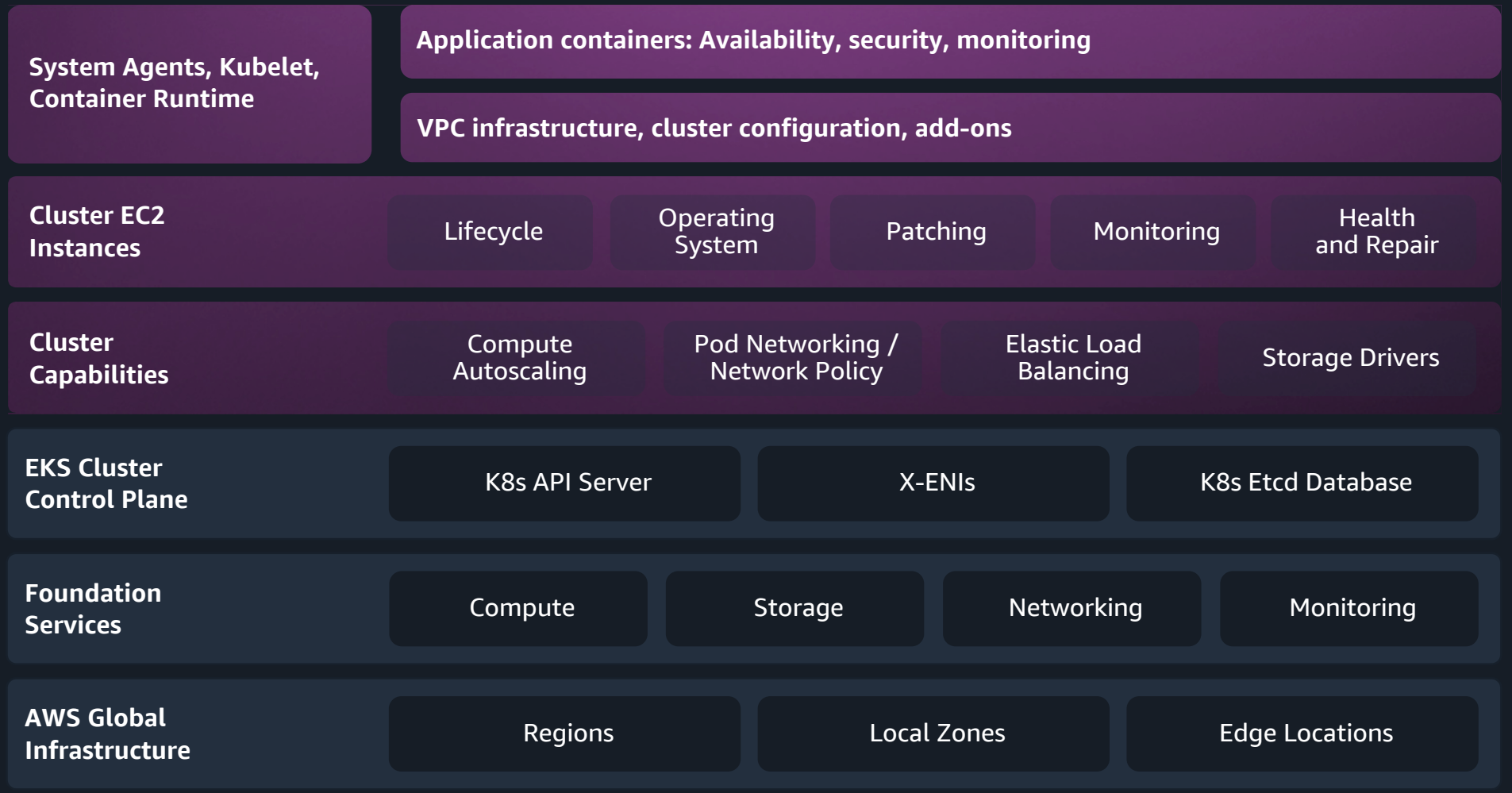
© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.

EKS Auto Mode cluster architecture



© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.

Shared Responsibility Model with EKS previously

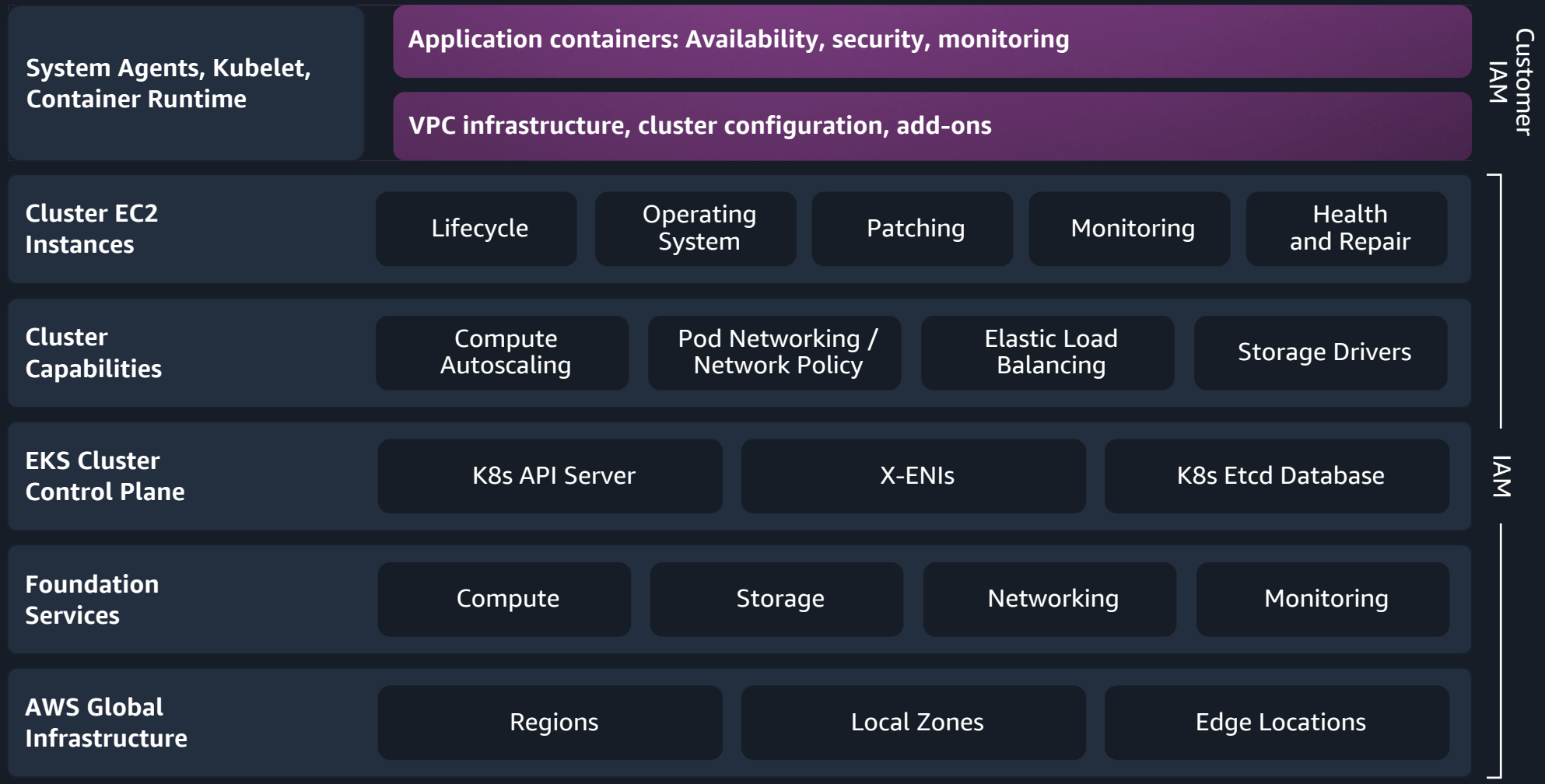


Managed by Customers

Managed by Amazon Web Services

© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.

Shared Responsibility Model with EKS Auto Mode



Managed by Customers

Managed by Amazon Web Services

© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.

EKS Auto Mode in action

© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.

WHO AM I?

Dan Levine



Working to enable Kubernetes & EKS in the Enterprise

Focused on the intelligent use of compute

© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.

What am I going to talk about?

- Capital One’s & Kubernetes
- Operating in a Federated Model
- Convergence on EKS Auto Mode
- Our Users

© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.

The Challenge: The Scattered Bottleneck

- **High SRE Effort:** Managing Kubernetes cluster and the underlying infrastructure required significant, specialized SRE bandwidth
- **Constant Churn:** A large Number of platform components need to be constantly updated, creating a large backlog
- **Scalability Issues:** Each team cannot effectively support the diverse needs of all the unique challenges they face, which stalls innovation



aws

© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.

The Solution: A Federated Model

- **Core Principle:** Shift from a purely centralized "do-it-for-you" model to a "do-it-with-you" collaborative framework.
- **Empowerment:** Designed to "meet platform teams where they're at" by providing them with the tools, standards, and autonomy they need.
- **Shared Ownership:** Enables a large group of people to come together to own, maintain, and evolve different parts of the platform.



aws

© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.

How It Works

- **What They Are:** Platform Teams and Special Interest Groups (SIGs)
- **How They Function:** Each SIG has a clear charter and ownership over a specific domain (e.g., SIG-Observability)
- They bring together experts and stakeholders to create, document, and maintain solutions for their domain
- They act as the primary point of contact and expertise for their components
- The central tooling team can continue to operate on evolution of Kubernetes



aws

© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.

The Residual Pain

Infrastructure Management

Persistent SRE overhead for provisioning, patching, and securing the underlying Kubernetes clusters and cloud infrastructure

Container Management

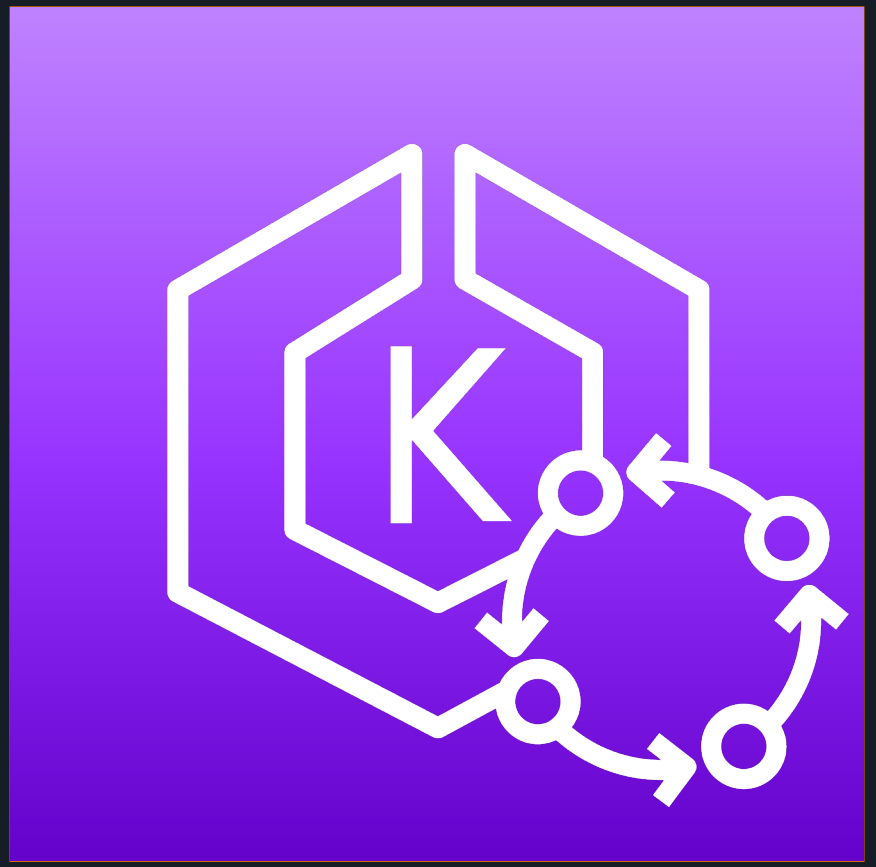
Ongoing operational load from image upgrading and container dependency management

aws

© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.

Here comes EKS Auto Mode

The next step in our intelligent usage of highly scalable compute



© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.

EKS Auto Mode is the Next Stage In Compute Evolution

Automated Infrastructure Management

EKS Auto removes the operational burden of provisioning, scaling, and patching the cluster's compute

Simplified Container Management

EKS Auto patches and manages the underlying containers that run common business functions

Happy Platform Teams!

aws

© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.

EKS Auto Mode 2025 features

© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.

FULLY-MANAGED EKS CLUSTERS

EKS Auto Mode 2025 Features

- US GovCloud and Local Zones support
- Advanced configuration
- SOCI parallel pull & unpack
- Security and compliance features
- Static and reserved capacity provisioning

aws

© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.

EKS AUTO MODE 2025 FEATURES

GovCloud (US) and Local Zones

Regional availability

EKS Auto Mode is available in all commercial regions where EKS is available, *except for AWS (China) Beijing and Ningxia regions*

In 2025, we expanded this to include the **AWS GovCloud (US) regions** and **AWS Local Zones**

© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.

EKS AUTO MODE 2025 FEATURES

Advanced Configuration

Flexibility without operational overhead

Use separate subnets and security groups for pods with `podSubnetSelectorTerms` and `podSecurityGroupSelectorTerms`

Advanced networking configurations like `associatePublicIPAddress` ,and `httpsProxy` / `noProxy`

Provide private certificate material to Auto Mode-managed EC2 instances with `certificateBundles`

aws

© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.

EKS AUTO MODE 2025 FEATURES

SOCI Parallel Image Pull & Unpack

Accelerate AI/ML workload startup

SOCI parallel container pull & unpack reduces Time-to-First-Token for AI inference workloads

Automatically enabled for GPU instances in EKS Auto Mode - G, P, Trn and other EC2 instances with local NVME storage

Zero configuration required optimized settings applied automatically

© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.

EKS AUTO MODE 2025 FEATURES

Security and Compliance Features

Delegate security and compliance to AWS

Encrypt both root and data storage volumes with `ephemeralStorage.kmsKeyID`

Enable FIPS-validated cryptographic modules in both EKS Auto Mode controllers and instances `advancedSecurity.fips`

Enable teams on Auto Mode without granting additional IAM permissions, just an `instanceProfile`

aws

© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.

EKS AUTO MODE 2025 FEATURES

Static and Reserved Capacity

AWS-Managed EC2 Reserved Capacity

Auto Mode can prioritize reserved capacity, increasing cost-efficiency using `capacityReservationSelectorTerms`. Supports both On Demand Capacity Reservations and Capacity Blocks for ML

For workloads that don't need Auto Mode's default dynamic and flexible compute, use Static Capacity to pre-provision capacity independently using `spec.replicas`

© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.

Wrapping up

© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.

EKS Auto Mode

Key Takeaways

Operational Transformation

Complex manual management to automated AWS-managed infrastructure

Focus on Innovation

Free teams to build business value, not manage infrastructure

Kubernetes Power + Simplicity

Full ecosystem access, less operational overhead

Enterprise Ready

Security, compliance, and performance without complexity

aws

© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.

Actions and resources



EKS Best Practices Guide



Getting Started with EKS



Hands-on Workshop Series

© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.



Thank You

Please complete the session survey in the mobile app